

MATH 4820 / Regression Analysis

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CHAPTER 4 MODEL ADEQUACY CHECKING



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4.1 INTRODUCTION

- Assumptions
 1. Relationship between response and regressors is linear (at least approximately).
 2. Error term, ε has zero mean
 3. Error term, ε has constant variance
 4. Errors are uncorrelated
 5. Errors are normally distributed (required for tests and intervals)

4.2 RESIDUAL ANALYSIS

- Definition of Residual (= data – fit):

$$e_i = y_i - \hat{y}_i$$

- Approximate average variance:

$$\frac{\sum (e_i - \bar{e})^2}{n - p} = \frac{\sum e_i^2}{n - p} = \frac{SS_{\text{Res}}}{n - p} = MS_{\text{Res}}$$

- Scaling helps in identifying **outliers** or extreme values
- **Four Methods**
 1. Standardized Residuals
 2. Studentized Residuals
 3. PRESS Residuals
 4. R-student Residuals

4.2.2 METHODS FOR SCALING RESIDUALS

1. Standardized Residuals

$$d_i = \frac{e_i}{\sqrt{MS_{\text{Res}}}}$$

- d_i 's have mean zero and variance *approximately* equal to 1
- Large values of d_i ($d_i > 3$) may indicate an outlier

2. Studentized Residuals

- MS_{Res} is only an approximation of the variance of the i th residual.
- Improve scaling by dividing e_i by the exact standard deviation:

$$\text{Var}(e_i) = \sigma^2(1 - h_{ii})$$

4.2.2 METHODS FOR SCALING RESIDUALS

2. Studentized Residuals

The studentized residuals are then:

$$r_i = \frac{e_i}{\sqrt{\hat{\sigma}^2(1 - h_{ii})}} = \frac{e_i}{\sqrt{MS_{\text{Res}}(1 - h_{ii})}}$$

- r_i 's have mean zero and unit variance.
- Studentized residuals are generally larger than the corresponding standardized residuals.

4.2.2 METHODS FOR SCALING RESIDUALS

3. PRESS Residuals

- Examine the differences: $y_i - \hat{y}_{(i)}$
 - [these are the differences between the actual response for the i th data point and the fitted value of the response for the i th data point, using all observations except the i th one.]
- Logic: If the i th point is unusual, then it can “overly” influence the regression model.
 - If the i th point is used in fitting the model, then the residual for the i th point will be small.
 - If the i th point is not used in fitting the model, then the residual will better reflect how unusual that point is.

4.2.2 METHODS FOR SCALING RESIDUALS

3. PRESS Residuals

- Prediction error: $e_{(i)} = y_i - \hat{y}_{(i)}$
- Calculated for each point, called PRESS residuals – [they will be used later to calculate the “prediction error sum of squares”].
- Calculate the PRESS residuals using

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}$$

4.2.2 METHODS FOR SCALING RESIDUALS

3. PRESS Residuals $Var[e_{(i)}] = Var\left[\frac{e_i}{1-h_{ii}}\right]$

– Variance:
$$= \frac{1}{(1-h_{ii})^2} \sigma^2 (1-h_{ii})$$
$$= \frac{\sigma^2}{(1-h_{ii})}$$

- The **standardized** PRESS residuals are

$$\frac{e_{(i)}}{\sqrt{Var(e_{(i)})}} = \frac{e_i / (1-h_{ii})}{\sqrt{\sigma^2 / (1-h_{ii})}} = \frac{e_i}{\sqrt{\sigma^2 (1-h_{ii})}}$$

- **Note:** these are the studentized residuals when MS_{Res} is used as the estimate of the variance.

4.2.2 METHODS FOR SCALING RESIDUALS

4. R-Student

- MS_{Res} is an “internal” estimate of variance
- Use a variance estimate that is based on all observations except the i th observation:

$$S_{(i)}^2 = \frac{(n-p)MS_{\text{Res}} - \frac{e_i^2}{(1-h_{ii})}}{n-p-1}$$

- The R -student residual is $t_i = \frac{e_i}{\sqrt{S_{(i)}^2(1-h_{ii})}}$
- This is an *externally studentized* residual.

DELIVERY TIME EXAMPLE

TABLE 4.1 Scaled Residuals for Example 4.1

Observation Number, i	$e_i = y_i - \hat{y}_i$ (1)	$d_i = e_i / \sqrt{MS_{Res}}$ (2)	$r_i = e_i / \sqrt{MS_{Res}(1 - h_{ii})}$ (3)	h_{ii} (4)	$e_{(i)} = e_i / (1 - h_{ii})$ (5)	$t_i = e_i / \sqrt{S_{(i)}^2(1 - h_{ii})}$ (6)	$[e_i / (1 - h_{ii})]^2$ (7)
1	-5.0281	-1.5426	-1.6277	0.10180	-5.5980	-1.6956	31.3373
2	1.1464	0.3517	0.349	0.07070	1.2336	0.3575	1.5218
3	-0.0498	-0.0153	-0.0161	0.09874	-0.0557	-0.0157	0.0031
4	4.9244	1.5108	1.5798	0.05838	5.2297	1.6392	27.3499
5	-0.4444	-0.1363	-0.1418	0.07501	-0.4804	-0.1386	0.2308
6	-0.2896	-0.0888	-0.0908	0.04287	-0.3025	-0.0887	0.0915
7	0.8446	0.2501	0.2704	0.08180	0.9198	0.2646	0.8461
8	1.1566	0.3548	0.3667	0.06373	1.2353	0.3594	1.5260
9	7.4197	2.2763	3.2138	0.49829	14.7888	4.3108	218.7093
10	2.3764	0.7291	0.8133	0.19630	2.9568	0.8068	8.728
11	2.2375	0.6865	0.7181	0.08613	2.4484	0.7099	5.9946
12	-0.5930	-0.1819	-0.1932	0.11366	-0.6690	-0.1890	0.4476
13	1.0270	0.3151	0.3252	0.06113	1.0938	0.3185	1.1965
14	1.0675	0.3275	0.3411	0.07824	1.1581	0.3342	1.3412
15	0.6712	0.2059	0.2103	0.04111	0.7000	0.2057	0.4900
16	-0.6629	-0.2034	-0.2227	0.16594	-0.7948	-0.2178	0.6317
17	0.4364	0.1339	0.1381	0.05943	0.4640	0.1349	0.2153
18	3.4486	1.0580	1.1130	0.09626	3.8159	1.1193	14.5612
19	1.7932	0.5502	0.5787	0.09645	1.9846	0.5698	3.9387
20	-5.7880	-1.7758	-1.8736	0.10169	-6.4432	-1.9967	41.5150
21	-2.6142	-0.8020	-0.8779	0.16528	-3.1318	-0.8731	9.8084
22	-3.6865	-1.1310	-1.4500	0.39158	-6.0591	-1.4896	36.7131
23	-4.6076	-1.4136	-1.4437	0.04126	-4.8059	-1.4825	23.0966
24	-4.5728	-1.4029	-1.4961	0.12061	-5.2000	-1.5422	27.0397
25	-0.2126	-0.0652	-0.0675	0.06664	-0.2278	-0.0660	0.0519

PRESS = 457.4000

R-CODE

```
delivery=read.csv("data-ex-3-1_(Delivery_Time).csv",header=T)
delivery.lm=lm(time~cases+dist,data=delivery)

##### residuals ei=yi-hat(yi)#####
res <- delivery.lm$residuals

##### standardized residuals #####
Res_MS=deviance(delivery.lm)/df.residual(delivery.lm)
stan_res<- delivery.lm$residuals/sqrt(Res_MS)

##### studentized residuals #####
stud_res <- rstandard(delivery.lm)

##### PRESS residuals #####
PRESS_res <- residuals(delivery.lm)/(1-lm.influence(delivery.lm)$hat)

##### standardized PRESS residual is #####
stanP_res <- PRESS_res/sqrt(Res_MS/(1-lm.influence(delivery.lm)$hat))
all.equal(stud_res, stanP_res)

##### R-student #####
library("MASS")
Rstu_res <- studres(delivery.lm)
```

4.2.3 RESIDUAL PLOTS

- Normal Probability Plot of Residuals
 - Checks the normality assumption
- Residuals against Fitted values, $\hat{y}_i$
 - Checks for nonconstant variance
 - Checks for nonlinearity
 - Look for potential outliers
- *Do not* plot residuals versus y_i (why?)

4.2.3 RESIDUAL PLOTS

Normal Probability Plot of Residuals

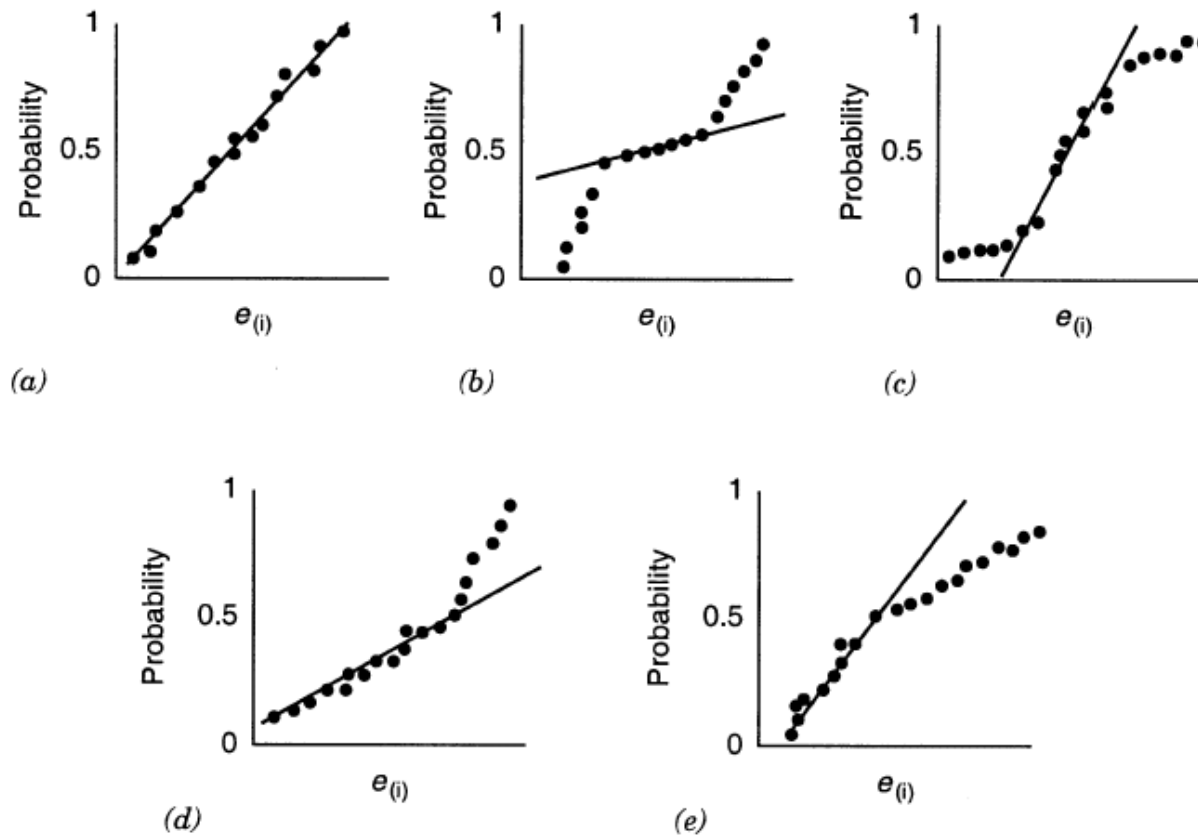


Figure 4.1 Normal probability plots: (a) ideal; (b) heavy-tailed distribution; (c) light-tailed distribution; (d) positive skew; (e) negative skew.

4.2.3 RESIDUAL PLOTS

Residuals against
Fitted values, $\hat{y}_i$

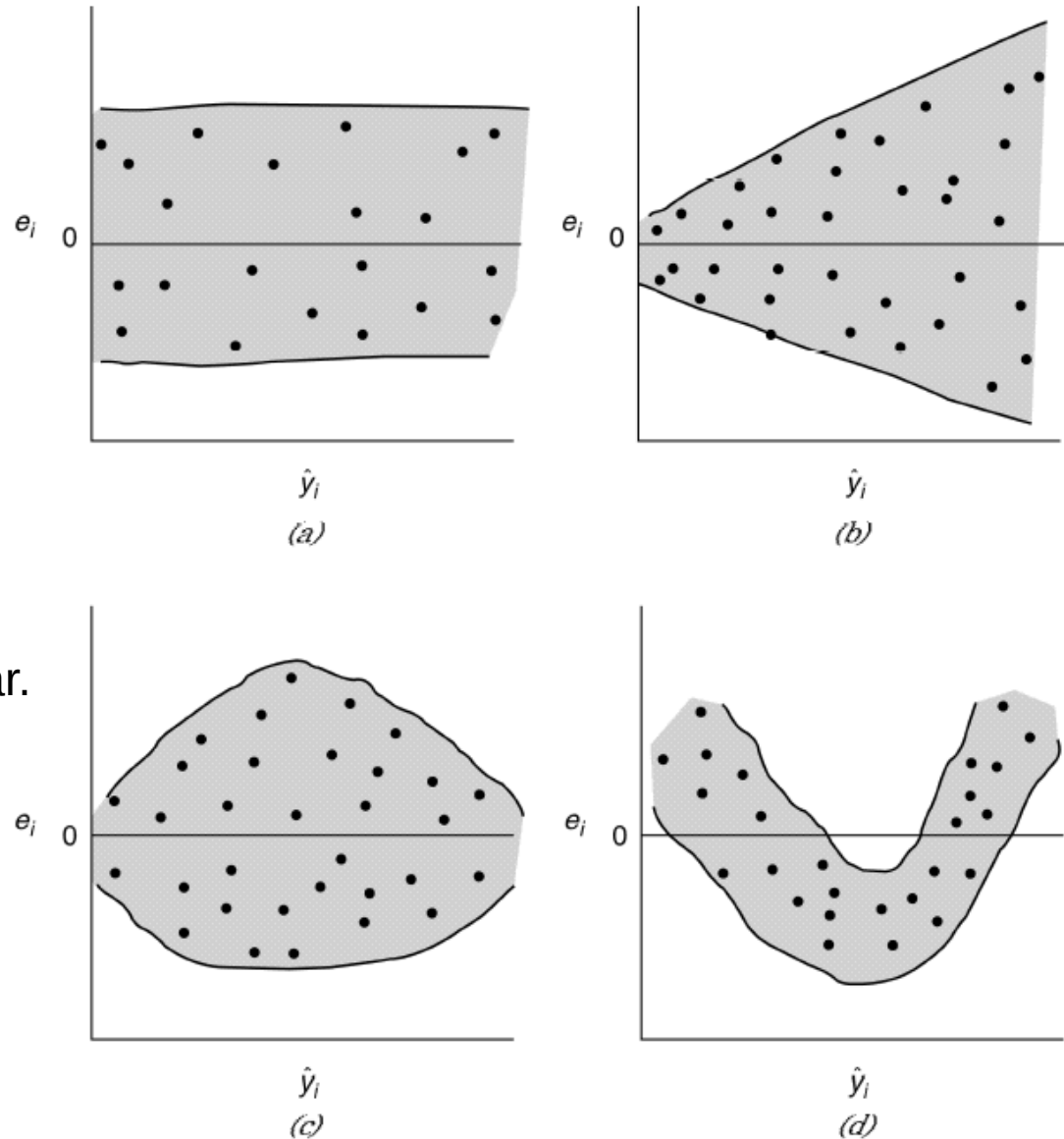


Figure 4.3 Patterns for residual plots:

a) satisfactory; b) funnel;
c) double bow; d) nonlinear.

EXAMPLE 4.4 THE DELIVERY TIME DATA

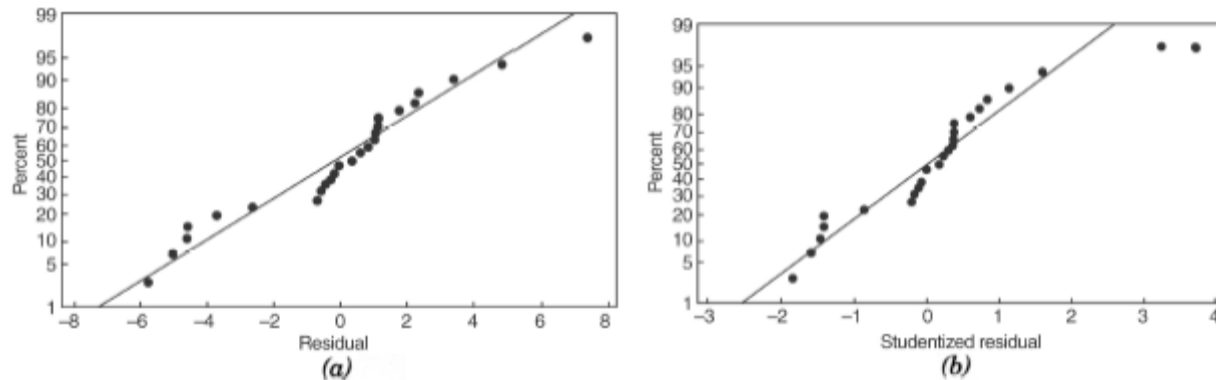


Figure 4.2 Normal probability plots of the residuals for the delivery time data: (a) ordinary least-squares residuals; (b) studentized residuals.

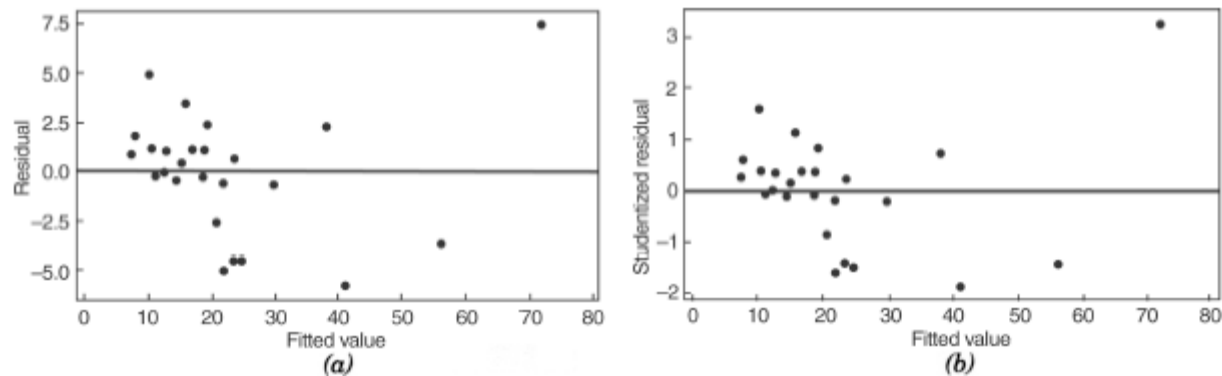


Figure 4.4 Plot of residuals versus predicted for the delivery time data: (a) original residuals; (b) studentized residuals.

4.2.3 RESIDUAL PLOTS

- Residuals against Regressors in the model
 - Checks for nonconstant variance
 - Look for nonlinearity
- Residuals against Regressors *not* in the model
 - If a pattern appears, could indicate that adding that regressor might improve the model fit.
- Residuals against time order
 - Check for Correlated errors

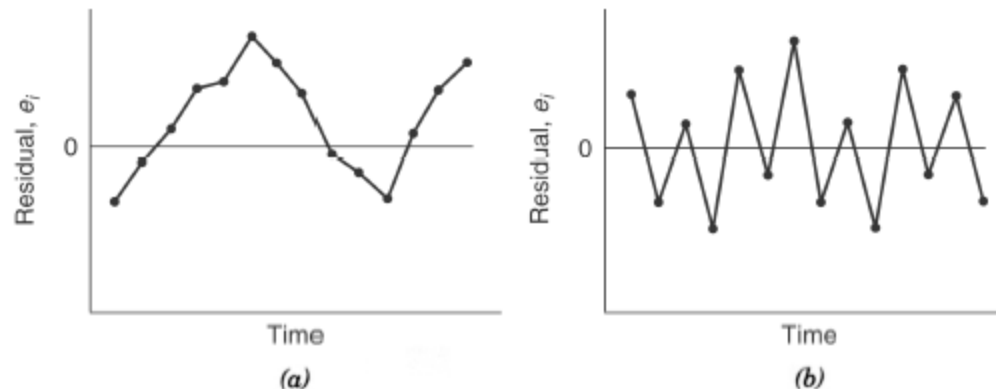


Figure 4.6 Prototype residual plots against time displaying autocorrelation in the errors: (a) positive autocorrelation; (b) negative autocorrelation.

EXAMPLE 4.4 THE DELIVERY TIME DATA

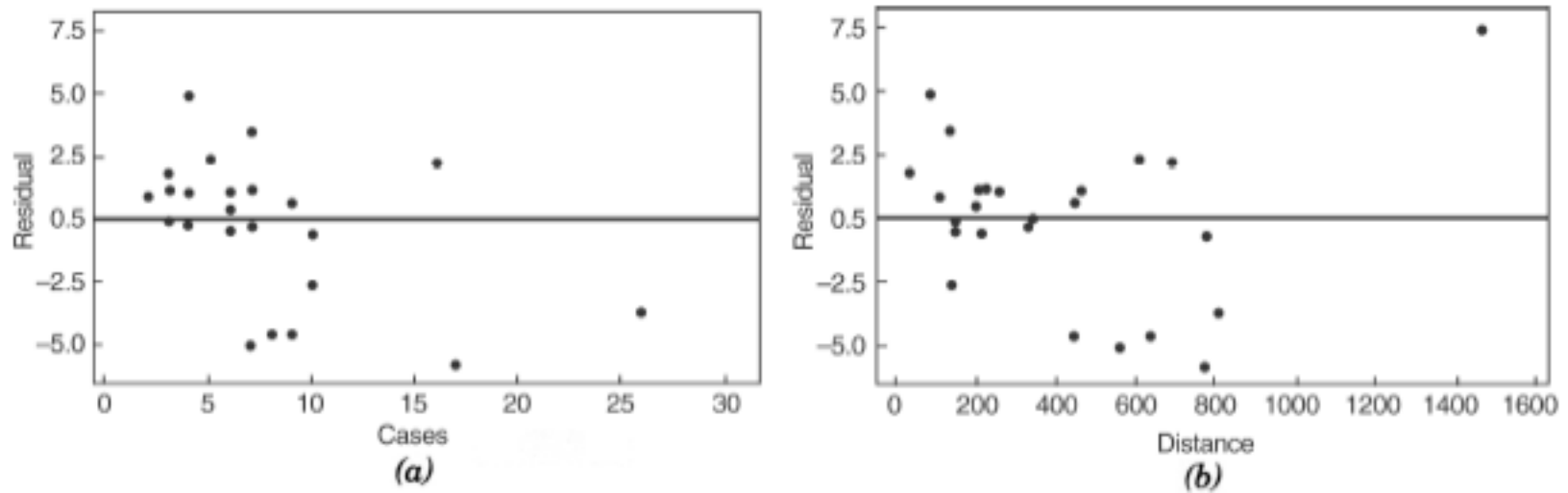


Figure 4.5 Plot of residuals versus the regressors for the delivery time data: (a) residuals versus cases; (b) residuals versus distance.

4.2.4 PARTIAL REGRESSION AND PARTIAL RESIDUAL PLOTS

Partial Regression Plots

- Why are these used?
 - To determine if the correct relationship between y and x_i has been identified
 - To determine the marginal contribution of a variable, given all other variables are in the model.

4.2.4 PARTIAL REGRESSION AND PARTIAL RESIDUAL PLOTS

Partial Regression Plots

- Method

Say we want to know the importance/relationship between y and some regressor variable, x_i .

- Regress y against all variables except x_i and calculate residuals
- Regress x_i against all other regressor variables and calculate residuals
- Plot these two sets of residuals against each other.

4.2.4 PARTIAL REGRESSION AND PARTIAL RESIDUAL PLOTS

Partial Regression Plots

- Interpretation
 - If the plot appears to be linear, then a linear relationship between y and x_i seems reasonable
 - If plot is curvilinear, may need x_i^2 or $1/x_i$ instead
 - If x_i is a **candidate variable**, and a horizontal “band” appears, then that variable adds no new information.

Example 4.5

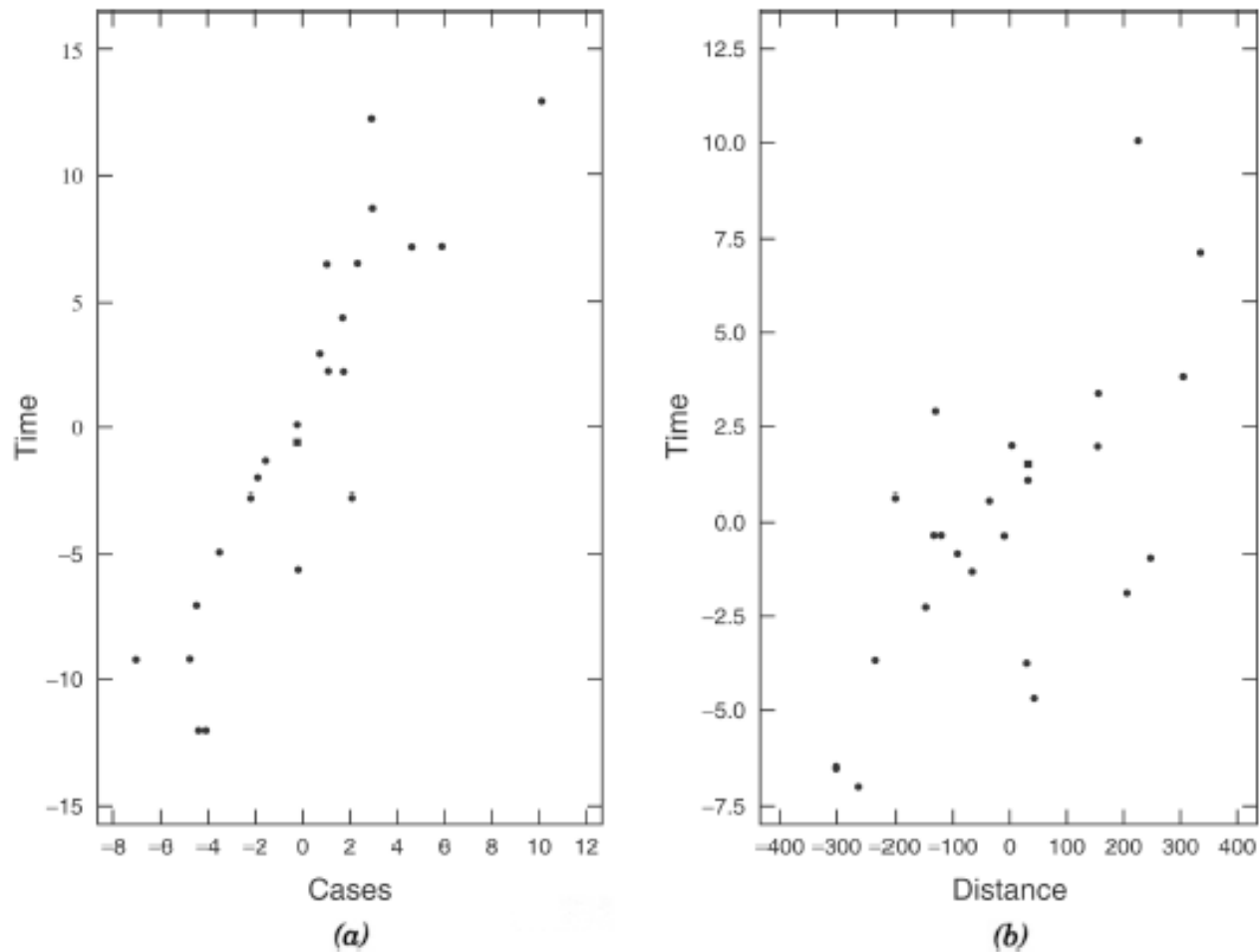


Figure 4.7 Partial regression plots for the delivery time data.

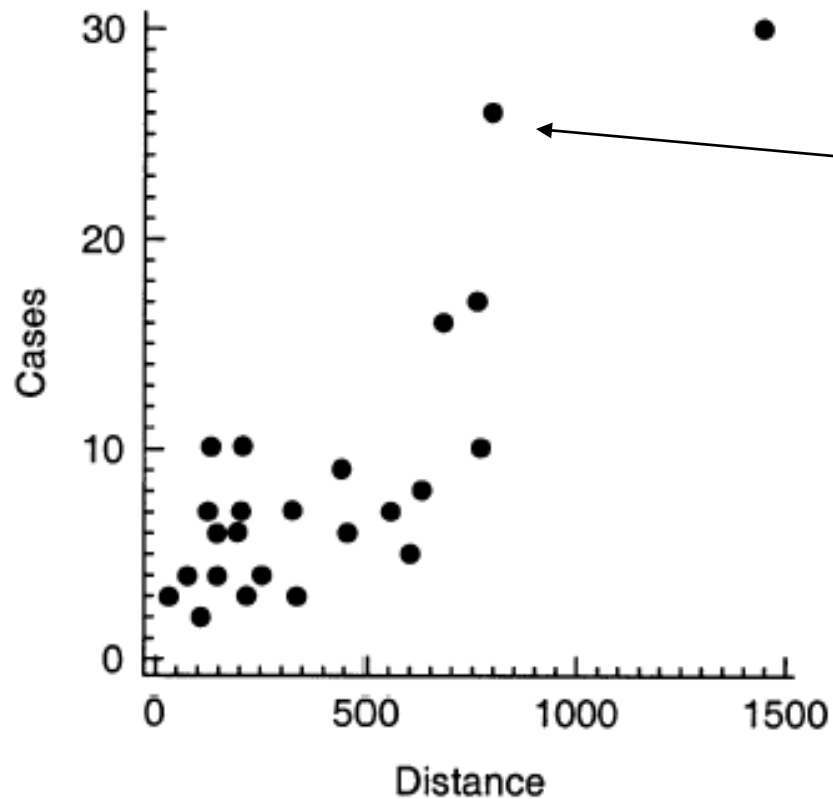
4.2.4 PARTIAL REGRESSION AND PARTIAL RESIDUAL PLOTS

Partial Regression Plots - Comments

- Use with caution, they only suggest possible relationships
- Do not generally detect interaction effects
- If multicollinearity is present, regression plots could give incorrect information
- The slope of the partial regression plot is the regression coefficient for the variable of interest!

4.2.5 OTHER RESIDUAL PLOTTING AND ANALYSIS METHODS

- Plotting regressors against each other can give info. about the relationship between the two:
 - may indicate correlation between the regressors.
 - may uncover remote points



Note location of these two point in the x - space

Figure 4.9 Plot of regressor x_1 (cases) versus regressor x_2 (distance) for the delivery time data in Table 3.2.

4.3 THE PRESS STATISTIC

- PRESS Residual:
- Prediction Error Sum of Squares (PRESS) Statistic:

$$e_{(i)} = y_i - \hat{y}_{(i)}$$

$$\begin{aligned} PRESS &= \sum (y_i - \hat{y}_{(i)})^2 \\ &= \sum \left(\frac{e_i}{1 - h_{ii}} \right)^2 \end{aligned}$$

- A small value of the PRESS Statistic is desired.
- See Table 4.1

4.3 THE PRESS STATISTIC

R^2 for Prediction Based on PRESS

$$R^2_{\text{prediction}} = 1 - \frac{PRESS}{SS_T}$$

- Interpretation:
 - We expect the model to explain about $R^2\%$ of the variability in prediction of a new observation.
- PRESS is a valuable statistic for comparison of models.

4.4 OUTLIERS

- An outlier is an observation that is considerably different from the others
- Formal tests for outliers
- Points with large residuals may be outliers
- Impact can be assessed by removing the points and refitting
- How should they be treated?

QUESTIONS?

- ANY QUESTION?

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

- Assumes
 - normality, independence, constant variance assumptions have been met.
 - Only the first-order or straight line model is in doubt.
- Requires
 - replication of y for at least one level of x .

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

- With replication, we can obtain a “model-independent” estimate of σ^2
- Say there are n_i observations of the response at the i th level of the regressor x_i , $i = 1, 2, \dots, m$
- y_{ij} denotes the j th observation on the response at x_i , $j = 1, 2, \dots, n_i$
- Total number of observations is $n = \sum_{i=1}^m n_i$

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

- Partitioning of the residual sum of squares:

$$SS_{\text{Res}} = SS_{\text{PE}} + SS_{\text{LOF}}$$

- SS_{PE} - pure error sum of squares
- SS_{LOF} - lack of fit sum of squares
- Note that the (ij)th residual can be partitioned, squared and summed.

$$y_{ij} - \hat{y}_i = (y_{ij} - \bar{y}_i) + (\bar{y}_i - \hat{y}_i)$$

$$(y_{ij} - \hat{y}_i)^2 = [(y_{ij} - \bar{y}_i) + (\bar{y}_i - \hat{y}_i)]^2$$

$$\sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_i)^2 = \sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2 + \sum_{i=1}^m n_i (\bar{y}_i - \hat{y}_i)^2$$

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

$$\underbrace{\sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \hat{y}_i)^2}_{\substack{SS_{\text{Res}} \\ DF=n-2}} = \underbrace{\sum_{i=1}^m \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2}_{\substack{SS_{\text{PE}} \\ DF=n-m}} + \underbrace{\sum_{i=1}^m n_i (\bar{y}_i - \hat{y}_i)^2}_{\substack{SS_{\text{LOF}} \\ DF=m-2}}$$

- If the assumption of constant variance is satisfied, then SS_{PE} is a “model-independent” measure of pure error.
- If the function really is linear, then $\hat{y}_i$ will be very close to $\bar{y}_i$ and SS_{LOF} will be quite small.

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

- Test Statistic:

$$F_0 = \frac{SS_{LOF} / (m - 2)}{SS_{PE} / (n - m)} = \frac{MS_{LOF}}{MS_{PE}}$$

- If $F_0 > F_{\alpha, m-2, n-m}$ conclude that the regression function is not linear. Why?

$$E(MS_{LOF}) = \sigma^2 + \frac{\sum_{i=1}^m n_i [E(y_i) - \beta_0 - \beta_1 x_i]^2}{m - 2}$$

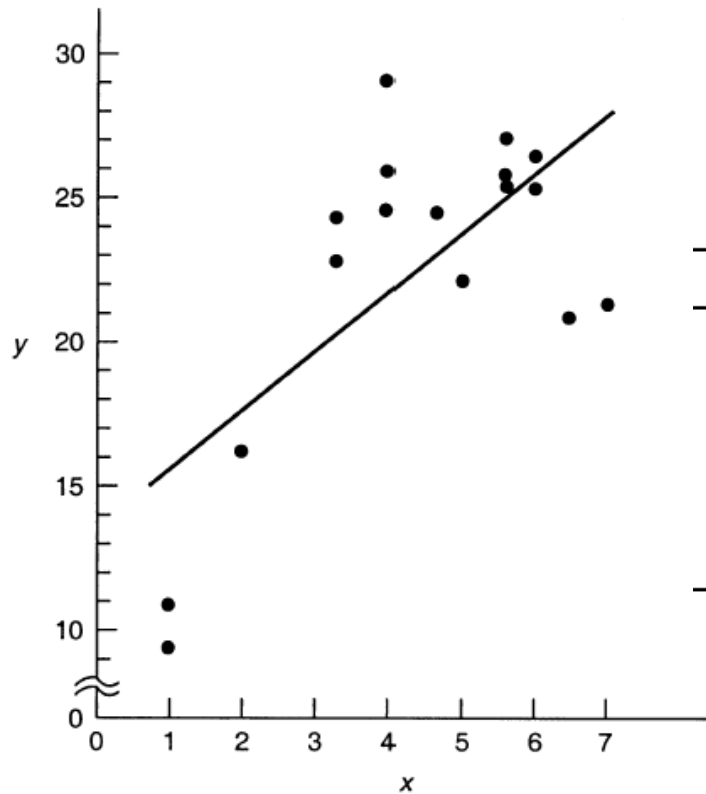
$$E(MS_{PE}) = \sigma^2$$

4.5 Lack of Fit of the Regression Model

A Formal Test for Lack of Fit

- If the test *indicates* lack of fit, abandon the model, try a different one.
- If the test indicates *no* lack of fit, then MS_{LOF} and MS_{PE} are combined to estimate σ^2 .

Example 4.8



Level of x	$\sum_j (y_{ij} - \bar{y}_i)^2$	Degrees of Freedom
1.0	1.1858	1
3.3	1.0805	1
4.0	11.2467	2
5.6	1.4341	2
6.0	0.6161	1
Total	15.5632	7

Figure 4.13 Data illustrating lack of fit of the straight-line model.

x	1.0	1.0	2.0	3.3	3.3	4.0	4.0	4.0	4.7	5.0
y	10.84	9.30	16.35	22.88	24.35	24.56	25.86	29.16	24.59	22.25
x	5.6	5.6	5.6	6.0	6.0	6.5	6.9			
y	25.90	27.20	25.61	25.45	26.56	21.03	21.46			

TABLE 4.2 Analysis of Variance for Example 4.8

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0	P Value
Regression	234.7087	1	234.7087		
Residual	252.9039	15	16.8603		
(Lack of fit)	237.3407	8	29.6676	13.34	0.0013
(Pure error)	15.5632	7	2.2233		
Total	487.6126	16			

An Approximate Procedure based on Estimating Error from Near-Neighbors

$$D_{ii'}^2 = \sum_{j=1}^k \left[\frac{\hat{\beta}_j(x_{ij} - x_{i'j})}{\sqrt{MS_{\text{Res}}}} \right]^2 \quad (4.27)$$

Pairs of points that have small values of $D_{ii'}^2$ are “near-neighbors,” that is, they are relatively close together in x -space. Pairs of points for which $D_{ii'}^2$ is large (e.g., $D_{ii'}^2 \gg 1$) are widely separated in x -space. The residuals at two points with a small value of $D_{ii'}^2$ can be used to obtain an estimate of pure error. The estimate is obtained from the range of the residuals at the points i and i' , say

$$E_i = |e_i - e_{i'}|$$

There is a relationship between the range of a sample from a normal population and the population standard deviation. For samples of size 2, this relationship is

$$\hat{\sigma} = (1.128)^{-1} E = 0.886 E$$

The quantity $\hat{\sigma}$ so obtained is an estimate of the standard deviation of pure error.

1. Compute the values of $D_{ii'}^2$ for all $n - 1$ pairs of points with adjacent values of $\hat{y}$. Repeat this calculation for the pairs of points separated by one, two, and three intermediate $\hat{y}$ values. This will produce $4n - 10$ values of $D_{ii'}^2$.
2. Arrange the $4n - 10$ values of $D_{ii'}^2$ found in 1 above in ascending order. Let E_u , $u = 1, 2, \dots, 4n - 10$, be the range of the residuals at these points.
3. For the first m values of E_u , calculate an estimate of the standard deviation of pure error as

$$\hat{\sigma} = \frac{0.886}{m} \sum_{u=1}^m E_u \quad (4.28)$$

Note that $\hat{\sigma}$ is based on the average range of the residuals associated with the m smallest values of $D_{ii'}^2$; m must be chosen after inspecting the values of $D_{ii'}^2$. One should not include values of E_u in the calculations for which the weighted sum of squared distance is too large.

See Example 4.10, pg. 162